

# SHAURYA VARSHNEY

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## EDUCATION

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**Indian Institute of Science Education and Research (IISER)** Bhopal, India  
*B.S. Engineering Sciences (Computer Science) | CPI: 7.31/10.0* 2023 – 2027  
– Relevant Coursework: Machine Learning, Computer Vision, Deep Learning, Intelligent Robotics, Multi-Agent RL, Data Structures & Algorithms

## RESEARCH & EXPERIENCE

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**Object Detection & Multi-Modal Distance Estimation for ADAS** 2025  
*Research Project* | Python, YOLOv11, OpenCV, Kalman Filter  
– Fine-tuned YOLOv11 for multi-class object detection achieving **94.2% mAP@0.5** in a real-time pipeline at **60+ FPS**.  
– Designed monocular depth estimation fusing pinhole camera models and ground-plane geometry, reducing distance error to **4.8%** across 50m — no stereo hardware.  
– Integrated Kalman Filtering for multi-object tracking, cutting identity switches by **35%** and enabling real-time collision-risk telemetry.

**HCLTech** Noida, India  
*Software Engineering Intern* | Node.js, ClickHouse, Docker, Vector May 2026 – Aug 2026  
– Architected a fault-tolerant log ingestion pipeline processing **50M+ daily telemetry events** across distributed microservices.  
– Replaced Logstash with Vector agents, cutting edge node CPU/memory by **40%**; built schema validation services at **12,000+ req/sec**.  
– Engineered ClickHouse sinks reducing storage by **65%** and analytical query latency to **150ms**.

**Transorg Analytics** Gurugram, India  
*Generative AI Intern* | LangChain, RAG, FAISS, FastAPI, Ollama May 2025 – Jul 2025  
– Built an end-to-end RAG pipeline parsing **5,000+ unstructured profiles** with a knowledge graph execution engine.  
– Improved semantic retrieval (Hit Rate@5) by **27%** via embedding fine-tuning, recursive chunking, and prompt engineering.  
– Reduced LLM token generation latency by **30%** through hardware-optimized Ollama orchestration via async FastAPI.

## TECHNICAL SKILLS

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**Languages:** Python, C++, SQL, JavaScript (ES6+), MATLAB  
**AI / ML:** PyTorch, LangChain, RAG, FAISS, Ollama, YOLOv11, OpenCV, Scikit-Learn, NumPy, Pandas  
**Backend & Databases:** FastAPI, Node.js, PostgreSQL, ClickHouse, Neo4j  
**Infrastructure:** Git, Docker, Vector, Unix/Linux

## LEADERSHIP

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**Co-Founder, Green Lungs Universal Foundation** — Scaled across 3 municipalities with 120+ volunteers; built digital ledger improving resource allocation by **22%**.  
**House Captain** — Led 40+ student cohort; organized 4 academic symposia.